The Normalized Opened-Atom Cap in the Literal Resonance Carrier: A Pointwise Factor Audit and Diagonal Transfer

Liang Wang
School of Artificial Intelligence and Automation
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

TPC-101 reduces the diagonal part of the positive multiplicative resonance carrier to two inputs: the resolved principal mass and a pointwise cap W_X for the normalized opened-atom weights. TPC-102 records $W_X \le X^{o(1)}$ as the first inexpensive gate after its three-ledger audit. We close that gate in the literal TPC-33–TPC-100 frame.

An invertible opened atom is $p = (\beta, z)$, with exact weight

$$w_p = |A_{\beta,X}|\,|\mathcal{W}_{\beta,X}(z)|\mu^2(\widetilde{D}_\beta(z))\mu^2(V_\beta(z)).$$

We pull this coefficient back through the lossless TPC-93 source–child inverse and place every factor in exactly one of six ledgers: physical rows, the opened logarithmic leg, the matched prefix, the row-gcd projector, exact content, and bounded local/smooth data. The literal row formula gives pointwise $O(\log X)$ coefficients; the logarithmic leg is $O(\log X)$; the matched prefixes, projector coefficients, and every remaining divisor expansion are bounded by a fixed power of the maximal divisor function on $X^{O(1)}$. No second column or fiber normalization occurs. Consequently, for a fixed integer A,

$$\boxed{W_X := \max_{p \text{ active}} w_p \ll (\log X)^A \left(\max_{n \le X^A} \tau(n)\right)^A = X^{o(1)}.}$$

The maximum is over the actual $q_X \nmid u_p$ support and retains one prescribed $h_0 \neq 0$, both polarizations, all source, content, projector, interval and mask labels, and the inherited physical normalization.

Combining the cap with the TPC-101 transfer shows that the width-weighted opened-atom diagonal ledger follows from the principal ledger; no separate diagonal-energy theorem is needed. This is a genuine L1 closure of Gate W, not an L2 arithmetic saving. The principal mass, cross-atom collisions, generic signed Möbius sector, full-block return, and strict physical endpoint remain open.

Keywords: opened atom; literal coefficient; pointwise cap; divisor bound; multiplicative resonance; fixed shift; provenance.

Literal-frame convention. All physical statements keep one prescribed $h_0 \in \mathbb{Z} \setminus \{0\}$, the actual active support, both polarizations, every native source and child label, and the unique inherited physical normalization. Nothing is specialized to $h_0 = 2$.

1 The first post-audit gate

At the critical TPC scales,

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)}, \qquad q_X \asymp Q_X.$$

After the conductor-one and $q_X \mid u$ branches have been returned, TPC-102 bounds the remaining positive resonant carrier by

$$\mathcal{C}^{\mathrm{abs}}_{\mathrm{exc},X} \ll X^{o(1)} Q_X^2 + B_X \left[\mathfrak{P}_X + \sqrt{W_X |\mathcal{H}_X| (q_X - 1) \mathfrak{P}_X} + \mathfrak{X}_X \right]. \tag{1}$$

Here $B_X \ll 1$ is the Wiener mass of the fixed low-window filter, $\mathfrak{P}_X$ is the principal resonance mass, $\mathfrak{X}_X$ is the width-weighted cross-atom excess, and W_X is the maximum opened-atom weight on the actual invertible support [2, 3].

The purpose of this paper is deliberately narrow: prove the pointwise bound

$$W_X \le X^{o(1)}$$

from the literal factor list. This is not a cancellation problem. It is nevertheless not a consequence of the row ℓ^1 bound: an abstract vector of ℓ^1-mass Q_X may place all its mass on one coordinate. The actual TPC row coefficient has a much stronger pointwise formula, and that formula must survive the entire source–child crosswalk.

2 The exact opened atom

2.1 Native record

Let

$$\beta = (\theta, c, \kappa), \qquad \theta = (L, \gamma, \ell, j, \sigma, v, \iota)$$

be a soluble, nonempty nonconstant branch in the canonical TPC-94 ledger. Put

$$b_\beta = c_\beta \kappa_\beta, \qquad g_\beta = (b_\beta, \sigma_\beta), \qquad B_\beta = b_\beta / g_\beta.$$

The resolved progression is

$$t_\beta(z) = \tau_\beta + B_\beta z, \qquad z \in I^*_\beta.$$

An r-free opened atom is

$$p = (\beta, z), \qquad u_p = U_\theta(t_\beta(z)), \qquad m_p = M_\theta(t_\beta(z)),$$

and it obeys the exact source identity

$$m_p j_\theta + h_0 = \sigma_\theta u_p. \tag{2}$$

We restrict to the remaining invertible support

$$1 \le N_\beta := \# I^*_\beta \le q_X, \qquad q_X \nmid u_p.$$

The restrictions are part of the actual support, not an ambient rectangular completion.

TPC-94 and TPC-100 define

$$A_{\beta,X} = c_{\theta,X} \mu(\kappa_\beta) \mu(B_\beta), \tag{3}$$
$$w_p = w_{\beta,z} = |A_{\beta,X}|\, |\mathcal{W}_{\beta,X}(z)| \mu^2(\widetilde{D}_\beta(z)) \mu^2(V_\beta(z)). \tag{4}$$

On canonical support the robustness content is one. Equation (4) is the exact positive atomization:

$$w_\beta = \sum_{z \in I^*_\beta} w_{\beta,z}.$$

There is no division by N_β, no cell-count normalization, and no second frequency factor [1, 6].

Definition 2.1 (Literal atom cap)**.** Let $\mathscr{P}^{\mathrm{inv}}_X$ be the preceding actual support after the single inherited normalization. Set

$$W_X = \begin{cases} \max_{p \in \mathscr{P}^{\mathrm{inv}}_X} w_p, & \mathscr{P}^{\mathrm{inv}}_X \ne \varnothing, \\ 0, & \mathscr{P}^{\mathrm{inv}}_X = \varnothing. \end{cases}$$

2.2 Why the word "normalized" introduces no new scalar

The physical units in (4) are exactly those of the TPC-93 low-window coefficient. TPC-93 explicitly reassembles the two polarized physical rows and states that no new fiber normalization is introduced. Thus "normalized opened atom" means an atom of the already normalized physical coefficient, not the probability weight $w_p/\sum w_p$. In particular, this paper does not insert a factor N_β^{-1}, Q_X^{-1}, or a columnwise renormalization.

3 A six-ledger pointwise factorization

The upstream coefficient contains many labels, but only finitely many kinds of factors. Pull p back through the explicit TPC-93 child-to-source inverse. Group its multiplicative factors as

$$w_p = |\mathsf{R}_p\, \mathsf{U}_p\, \mathsf{H}_p\, \mathsf{V}_p\, \mathsf{C}_p\, \mathsf{S}_p|. \tag{5}$$

This is a partition of the factors already present in (4), with the following meaning.

Ledger	Literal content	Pointwise envelope
R_p	The two physical row coefficients $\gamma^{(1)}_{\alpha_p}\gamma^{(2)}_{\gamma_p}$	$\ll (\log X)^2$
U_p	The opened coefficient $-\mu(u_p)\log u_p$ on the moving leg	$\ll \log X$
H_p	The matched half-prefix on the opposite leg and any already evaluated residual prefix	$X^{o(1)}$
V_p	The row-gcd projector coefficient and its coprimality indicator	$\le \tau(v_p)$ in modulus
C_p	$\mu(\kappa_\beta)\mu(B_\beta)$, squarefree indicators, and the two exact-content extraction masks	≤ 1 in modulus
S_p	Fixed-period, residue, smooth, Mellin/BV, interval, and actual support factors, including the outer active-divisibility indicator and the location of the single global normalization	$X^{o(1)}$ pointwise

Table 1: Every source factor is charged exactly once. The table does not replace the full provenance record by six anonymous numbers; it only groups factors for a pointwise estimate.

More explicitly, this is a crosswalk to the upstream formula, not a new schematic coefficient. The factors inside $c_{\theta,X}$ and $\mathcal{W}_{\beta,X}(z)$ are assigned to $\mathsf{R}_p, \mathsf{U}_p, \mathsf{H}_p, \mathsf{V}_p, \mathsf{S}_p$; $\mu(\kappa_\beta)\mu(B_\beta)$ and the two squarefree extraction factors are assigned to C_p; and the active-divisibility indicator and inherited normalization are recorded once in S_p. Unit phases disappear upon taking the modulus and are not additional positive-weight factors.

Lemma 3.1 (Physical row factors)**.** *Uniformly on the actual dyadic packet,*

$$|\mathsf{R}_p| \ll_{\mathscr{D}} (\log X)^2.$$

Proof. The literal TPC-25/TPC-33 row formula is

$$\gamma^{(i)}_\alpha = \mu(d_\alpha)(\log \ell_\alpha)\omega^{(i)}_D(d_\alpha)\psi^{(i)}_L(\ell_\alpha/L)\zeta^{(i)}_\alpha.$$

The final three factors are fixed and bounded on the physical packet, while $\ell_\alpha \le X^{O(1)}$. Hence each row coefficient is $O_{\mathscr{D}}(\log X)$ [4, 5]. In fact TPC-25 records directly $\|\gamma^{(i)}\|_\infty \ll \log(2L)$. Both polarizations use the same estimate. □

Lemma 3.2 (Divisor-type factors)**.** *There are fixed constants $A_0, C_0 > 0$, depending only on the dyadic data and the fixed decomposition, such that*

$$|\mathsf{U}_p\mathsf{H}_p\mathsf{V}_p\mathsf{C}_p\mathsf{S}_p| \ll (\log X)^{A_0}\Delta_X^{A_0}, \qquad \Delta_X := \max_{1\le n\le X^{C_0}} \tau(n).$$

Proof. All effective targets, divisors, row coordinates, and content variables are at most X^{C_0} for one fixed C_0. The opened logarithmic coefficient is at most $O(\log X)$. TPC-98 records the uniform bounds

$$\sum_{\substack{u|N \\ T_X<u\le U_{0,X}}} |\mu(u)|\log u = X^{o(1)}$$

and

$$|\mathsf{H}_m(j)| + |\mathsf{A}_{m,T_X}(j)| + |\mathsf{A}_{m,U_{0,X}}(j)| = X^{o(1)}$$

on the literal support [7]. More explicitly, these are bounded by a fixed product of powers of $\log X$ and divisor functions of effective integers.

For the projector,

$$\lambda_{G_X^{\mathrm{row}}}(v) = \sum_{\substack{d|v \\ d\le G_X^{\mathrm{row}}}} \mu(v/d),$$

so $|\lambda_{G_X^{\mathrm{row}}}(v)| \le \tau(v)$. Every Möbius factor, squarefree indicator, and coprimality mask has modulus at most one. The fixed smooth and residue factors are bounded. The absolutely convergent separation representation has bounded total projective mass, so each of its retained coefficient atoms is bounded by that mass. The number of multiplicative factor types is fixed, giving the displayed envelope. □

4 Closure of Gate W

Theorem 4.1 (Normalized opened-atom cap)**.** *For the actual TPC-100 invertible opened-atom support, with the literal TPC-33–TPC-94 factors and the single inherited physical normalization,*

$$\boxed{W_X \ll_{\mathscr{D},h_0} (\log X)^A \left(\max_{n\le X^A} \tau(n)\right)^A = X^{o(1)}} \tag{6}$$

for one fixed $A > 0$.

Proof. If the support is empty there is nothing to prove. Otherwise apply lemmas 3.1 and 3.2 to the exact partition (5). The classical divisor estimate, uniformly for $n \le X^A$, is

$$\tau(n) \le \exp\left(O_A\left(\frac{\log X}{\log\log X}\right)\right) = X^{o(1)}.$$

A fixed product of this quantity and powers of $\log X$ remains $X^{o(1)}$. No count over rows, children, contents, or branches occurs in a pointwise maximum. □

Remark 4.2 (Uniformity and the fixed shift)*.* The exponent $o(1)$ is uniform over the fixed dyadic packet data and the one prescribed h_0. The result is not an average over shifts. The condition $q_X \nmid u_p$ merely selects the actual invertible support and does not enlarge the maximum.

Corollary 4.3 (The diagonal ledger is no longer independent)**.** *Let*

$$\mathfrak{P}_X = \sum_H \frac{2H}{q_X - 1} M_H^{\mathrm{inv}}.$$

Then the TPC-101 principal-to-diagonal transfer becomes

$$\sum_H g_{q_X}(H)\sqrt{\mathfrak{D}_H^\circ} \le X^{o(1)}\sqrt{|\mathcal{H}_X|(q_X-1)\mathfrak{P}_X}.$$

Consequently, if

$$\mathfrak{P}_X \le X^{o(1)} Q_X^2,$$

then

$$\sum_H g_{q_X}(H)\sqrt{\mathfrak{D}_H^\circ} \le X^{o(1)} Q_X^2.$$

Proof. Insert theorem 4.1 into the exact TPC-101 transfer

$$\sum_H g_{q_X}(H)\sqrt{\mathfrak{D}_H^\circ} \le \sqrt{W_X|\mathcal{H}_X|(q_X-1)\mathfrak{P}_X}.$$

Use $|\mathcal{H}_X| \le (q_X-1)/2$ and $q_X \asymp Q_X$. □

Remark 4.4 (What was and was not removed). The corollary removes the need for a separate positive opened-atom diagonal theorem. It does not prove the principal bound, and it does not touch the distinct-atom cross ledger $\mathfrak{X}_X$. In particular, same-branch different-z atoms still collide in that cross ledger.

5 Why an ℓ^1 row bound was insufficient

Proposition 5.1 (Sharp abstract obstruction)**.** *Knowledge only of*

$$\sum_\alpha |\gamma_\alpha^{(i)}| \le Q_X$$

cannot imply a subpower opened-atom cap. Indeed, there are nonnegative abstract row families satisfying this inequality for which one two-row atom has weight Q_X^2.

Proof. Take one active row in each polarization and give both rows coefficient Q_X; set all other row coefficients to zero and all local factors to one. Each ℓ^1 bound is saturated, while the product atom has weight Q_X^2. □

Remark 5.2 (The counterexample is not physical). Proposition 5.1 does not contradict theorem 4.1: it deliberately discards the explicit TPC-33 row formula. It explains why TPC-101 and TPC-102 correctly isolated Gate W instead of inferring it from an aggregate norm.

6 Finite provenance regression

The deterministic script `experiments/tpc103_atom_cap_audit.py` performs three kinds of exact checks:

1. the row-gcd projector identity

$$\sum_{v|(d,e)} \sum_{\substack{g|v \\ g\le G}} \mu(v/g) = \mathbf{1}_{(d,e)\le G};$$

2. the source-to-child and child-to-source round trip on both polarizations for finite primitive samples; and
3. a schema audit that assigns every literal factor category to exactly one of the six ledgers in (5).

The committed JSON records the counts and the claim boundary. These are regression tests for finite identities; they are not numerical evidence for an asymptotic Möbius estimate. The asymptotic input in theorem 4.1 is only the standard divisor bound.

7 Route consequence and claim boundary

Proved here

- The exact opened atom has a six-ledger factor audit with no duplicated normalization.
- Every literal pointwise factor is logarithmic, bounded, or divisor-function sized on $X^{O(1)}$.
- The actual normalized cap satisfies $W_X = X^{o(1)}$.
- Therefore the TPC-101 width-weighted diagonal gate follows from the unresolved principal gate.

The finite factor partition is L0; its lossless attachment to the actual growing fixed-h_0 carrier and the resulting cap are L1 bookkeeping progress.

Not proved here

- No bound for $\mathfrak{P}_X$ or $\mathfrak{X}_X$ is proved.
- No positive-resonance closure, generic affine Möbius cancellation, determinant lower bound, zero-mode saving, high-frequency return, or TPC-18 full-block cover is proved.
- No fixed-power saving for the complete growing fixed-h_0 coefficient is obtained; hence no L2 event occurs.
- No Hardy–Littlewood asymptotic, prime-pair lower bound, parity breakthrough, or twin-prime conclusion is proved or implied.

The strict physical endpoint remains

$$\Lambda_{\text{phys}} < \frac{1}{400}.$$

The present subpower cap spends no fixed exponent in that ledger. The next gate is the resolved principal-mass disintegration; the cross-map quotient comes only after that positive principal term is understood, in accordance with TPC-MVP1 [8].

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